STAMODIAAS: A Stage Model for Asset Administration Shell Integration on the Shop Floor

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Abstract

The Asset Administration Shell (AAS) is increasingly used to support digitalization on the shop floor, yet companies still lack a practical roadmap for introducing AAS into internal production processes. This paper presents STAMODIAAS (*Stage Model for Integrating the AAS on the Shop Floor*), a five-stage model for AAS integration on the shop floor. For each stage, the required infrastructure, competencies, resources, implementation steps, and effort are described. In addition, representative AAS-based use cases are classified and mapped to the stages to support self-evaluation of the current digitalization level and to identify suitable next steps and business potentials. STAMODIAAS is evaluated in academic demonstrators and further validated using practical insights from AUDI AG. The results show that early stages, such as infrastructure setup and asset identification, can be realized with comparatively low effort, whereas advanced submodels and live data integration require significantly stronger competencies in connectivity and IT/OT integration. The model therefore provides a reference for staged AAS studies and helps with decision making in modern shopfloor automation.

1 Introduction

The Asset Administration Shell (AAS) is an information model that provides a single source of truth for integrating and managing data across the lifecycle of physical and non-physical assets in manufacturing systems [1]. By modeling assets as digital entities, the AAS enables an asset-centric approach to data management, which forms the basis for a wide range of data integration and interoperability use cases. Building on this concept, initiatives such as Manufacturing-X[1] extend AAS capabilities to support cross-company data exchange and collaboration.

However, continuous research and development have significantly expanded the AAS feature set, resulting in increased complexity when introducing the concept into industrial environments. To implement the AAS in practice, companies must make a range of architectural and organizational decisions, including the choice of AAS type [2], level of reactiveness, repository strategy, or modeling depth. These decisions extend further to the selection of tools and software stacks, as well as the definition of data governance and ownership processes. At the same time, a fragmented landscape of tools, architectural approaches and application scenarios poses additional challenges for companies aiming to deploy the AAS in productive settings.

The breadth and interdependence of these design choices contribute significantly to the perceived complexity of AAS adoption. As a result, practitioners are often confronted with a wide solution space but lack structured guidance, particularly with regard to implementation effort, timelines, and associated costs [3].

This paper provides a practical multi-stage model for companies seeking to adopt the AAS in their organization. Rather than focusing on theoretical concepts, it addresses the key questions of how the AAS can be realistically implemented and integrated into existing company processes, what efforts in terms of tools, capabilities, skills are required, and which use cases are most relevant for industrial applications. The identified use cases are evaluated based on their implementation complexity and required resources, then arranged into a structured sequence from simple to complex. The result is a practical multi-stage roadmap and procedure model that de-risks AAS adoption, supports self-evaluation, and enables informed capacity and budget planning.

The staged model focuses on a plant-level rollout, focuses on existing brownfield manufacturing setups. The discus-

[1] https://www.plattform-i40.de/IP/Navigation/DE/Manufacturing-X/Initiative/initiative-manufacturing-x.html

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sion is vendor and tool agnostic and neutral regarding software architecture. Dataspace interoperability is out of scope. In the remainder of the paper, Section 2 introduces the fundamentals, while Section 3 highlights related studies. Section 4 then outlines STAMODIAAS (*Stage Model for Integrating the AAS on the Shop Floor*). Section 5 presents the application of STAMODIAAS in two academic demonstrators and its validation in the automotive domain, discussing the results and providing recommendations. Finally, Section 6 concludes the paper and outlines opportunities for future research.

2 Fundamentals

In the following, we discuss the foundational aspects of shop floor digitization and the AAS.

2.1 Current Approaches in Shop Floor IT

Digitizing shop floors and production systems refers to the integration of production assets with digital information systems to increase adaptability and efficiency, and to enable various use cases such as condition monitoring or predictive maintenance [4]. In the context of Industry 4.0 and Manufacturing-X, machines and automation systems become connected data providers and consumers: information is integrated vertically from individual assets up to networks, and horizontally across shop floors, or even beyond company boundaries, to support collaborative manufacturing ecosystems [5].

In industrial reality, this transformation predominantly takes place in brownfield environments. Existing machines, automation systems, and IT solutions must continue operating, requiring incremental digitization and coexistence of legacy OT with modern IT. Heterogeneous semantics, proprietary interfaces, real-time constraints, and high availability requirements pose significant challenges. In contrast, existing production environments are typically designed in a system-centric architecture, meaning that a manufacturing cell, line, or machine is treated as a closed, self-contained unit that is automated and operated independently. This paradigm stands in contrast to the asset-centric architectures, where each asset is digitally represented and interconnected across the value chain (**Figure 1**). Such architectures form the foundation for unlocking the full operational benefits of digital transformation.

2.2 Asset Administration Shell

The AAS is a standardized digital representation of an asset such as a machine, production process, or software [1]. It provides a uniform, technology-neutral way to structure and exchange information and services about assets [6].

An AAS is composed of elements such as an AAS identifier, the asset ID, and a description of the asset. Modular submodels contain further use-case-specific information [1]. Common examples of submodels include the Digital Nameplate (basic information), Technical Data (key specifications), Handover Documentation (manuals and service reports), Bill of Material, Capabilities (what the asset can do), and Time-Series Data (operational signals). The AAS can be extended with further submodels depending on the application context [7].

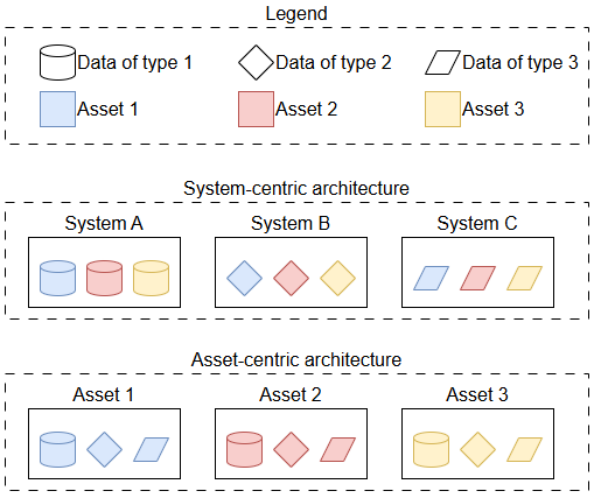


Figure 1: System-centric vs. Asset-centric information architecture

Submodels and their elements can additionally carry semantic identifiers that link them to standardized templates and dictionaries. For practical on-site access, the physical asset can be tagged with a machine-bound QR code that encodes (or resolves to) the relevant asset and AAS identifiers so that the corresponding AAS and its submodels can be retrieved quickly [8].

From a data integration perspective, an AAS is typically exposed via standard HTTP/REST services [9]. A Registry maps AAS or Submodel IDs to concrete endpoints and a Discovery service accelerates searches [10]. Those endpoints provide the access to the AAS itself, its submodel references, and each Submodel Interface.

3 State of the Art

Building on practical shop floor challenges and AAS fundamentals, the following section reviews the state of the art in AAS-related methods, architectures, and rollout approaches.

The literature highlights the AAS across diverse industrial production use cases. It primarily functions as an information exchange and integration layer for shop floor assets, enabling identification, access to lifecycle data, and integration with IT systems [11]. Case studies further demonstrate its application in maintenance, plug-and-produce, orchestration, and condition monitoring [12]. Moreover, the AAS provides a foundation for agent-based production systems, enabling autonomous asset interactions and increased manufacturing flexibility [13].

While these and numerous other studies focus on implementing specific use cases based on the AAS, other contributions address the systematic deployment of AAS technology within production systems, including architectural integration and migration strategies. [14] recommends defining shell and asset identifiers, starting with a minimal

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set of generic submodels (Index, Property Value Statement, Documentation, Communication), then adding asset-specific submodels and mapping the information model to AutomationML with OPC UA interfaces for shop floor integration, thereby outlining a structured implementation of AAS in a manufacturing environment. [15] proposes a use-case-driven pipeline: capture requirements in a conceptual model, separate concerns to decide which submodels and relationships are needed, assign semantic identifiers from existing vocabularies, and validate by wiring real data sources (e.g., OPC UA/SQL). [16] complements this with deployment guidance, treating submodels as deployable units and selecting hosting, execution, communication, and reachability options via the Deployment–Execution–Communication–Reachability scheme, with rules and scenario-based patterns suited to shop floor constraints.

Architectural work increasingly treats the AAS as the canonical semantic layer in production IT. [17] shows how PLM/ALM data (PLM-XML and ReqIF) can be transformed into dedicated submodels and linked via RelationshipElement inside AAS packages to achieve model-level traceability across lifecycle tools. [5] extends this to cross-company settings by combining AAS servers with OPC UA servers and a dataspace connector for sovereign exchange within a reference Manufacturing-X architecture. [18] introduces an AAS-centred middleware that maps heterogeneous robot and machine interfaces and web services into an AAS information model, demonstrating low-latency coordination on a modular assembly station. [19] reports registry-based orchestration of heterogeneous robots using MQTT events for fast signalling and REST requests for structured queries, and highlight needs like edge deployment and authorisation.

Across applications, several works describe practical paths for bringing AAS into existing production environments. [20] implement the AAS as a standardized middleware layer around machines to expose data and services through a uniform interface. [21] shows onboarding and integration approaches based on AASX exchange, low-code import/export, and digital connectivity across enterprise systems. [22] contributes a brownfield retrofit approach in which Time Series Data and Service Request Notification submodels connect operational data with maintenance processes. [23] extends this towards remote operation by using an AAS data space architecture for event-based interaction between production lines and external operators.

Despite this breadth, what is still missing is a plant-level, stepwise rollout method for AAS on the shop floor, one that links typical machine/line use cases to maturity stages and states required infrastructure, competencies, resources, and effort per stage. Existing guidance on AAS backend integration consolidates patterns and high-level recommendations but stops short of scenario-level effort baselines or competency matrices [24]. Companion materials and showcases demonstrate interoperability in testbeds, yet they are not structured as a reusable implementation method for factories and rarely report comparative effort or capability needs [25]. This paper addresses that gap with STAMODIAAS, a staged model that provides a minimal viable stack, stepwise progression, and indicative effort and competency estimates.

4 STAMODIAAS

In this section, we present STAMODIAAS, a five-stage model for AAS integration on the shop floor. The stages and their associated roles are summarized in **Table 1**, followed by representative production use cases mapped to each stage.

4.1 Stage 1: Infrastructure Groundwork

Stage 1 (S1) focuses on infrastructure, beginning with the make-or-buy decision, which weighs the benefits of developing a custom solution using open-source frameworks against licensing a proprietary one. Subsequently, the IT solution architecture is designed, defining core components such as AAS servers, AAS registries for discovery, and the underlying network infrastructure. A critical choice is the infrastructure deployment model, whether on-premise, private cloud, or public cloud, to balance data sovereignty, availability, and scalability. Finally, a concept for software operation is established to ensure stable and secure long-term performance.

The successful execution of this stage requires capabilities in strategic business and technical analysis to facilitate an evidence-based make-or-buy decision, predicated on comprehensive financial and architectural evaluations. Furthermore, advanced proficiency in solution and cloud architecture is required to design a resilient IT/OT infrastructure that systematically addresses requirements for data sovereignty and security within the selected deployment model. Ultimately, the establishment of a sustainable operational concept is contingent upon specialized skills in DevOps and cybersecurity to ensure automated, secure, and reliable system performance throughout its lifecycle.

4.2 Stage 2: Asset Identification

The subsequent stage (S2) focuses on the initial digital representation of physical components within the production environment. This process involves the systematic identification and cataloging of relevant shop floor assets. For each identified asset, a corresponding AAS is instantiated, which is assigned a globally unique identifier but is intentionally created without any submodels. The crucial final step is the establishment of a physical-to-digital link, which is typically realized by a machine-readable carrier, such as a QR code containing the unique ID, directly to the physical asset.

The execution of this stage requires a combination of domain-specific and technical competencies. Foundational knowledge of the production environment and manufacturing processes is essential for the accurate identification and selection of relevant physical assets. Competence in operating the designated AAS platform is necessary to perform the technical task of instantiating the shells and generating their unique identifiers. Furthermore, skills in process man-

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agement and logistical coordination are required to systematically implement the physical-to-digital linkage, ensuring the correct association between the digital ID and its corresponding physical counterpart on the shop floor.

4.3 Stage 3: Basic Submodels

While no formal specification distinguishes between basic and advanced submodels, a practical differentiation arises from their implementation complexity and functional scope. Basic submodels primarily serve a descriptive purpose, encapsulating static or infrequently changing data. Typical examples of this category include submodels representing an asset's nameplate, its technical documentation, or its master data. Basic submodels are broadly deployed across most assets, whereas advanced submodels are reserved for a specific subset of assets, determined by the requirements of individual use cases.

Implementing use cases with basic submodels requires domain knowledge to identify relevant asset information and skills to source this data. The Data Engineer also needs data modeling skills and a basic understanding of AAS specifications and available standard submodels.

4.4 Stage 4: Advanced Submodels

Advanced submodels primarily serve an operational purpose, encapsulating dynamic, real-time data and exposing interactive capabilities such as operations and events. Typical examples of this category include submodels representing an asset's condition monitoring, process control, or predictive maintenance data. Advanced submodels are selectively deployed across a limited subset of assets, driven by the specific requirements of individual use cases rather than broadly applied across the entire asset landscape.

Implementing use cases with advanced submodels requires deep technical expertise in IT/OT integration to establish live data connections between the physical asset and its digital representation. Additionally, proficiency in AAS specifications is essential, particularly regarding the modeling of operations, events, and dynamically updated properties. Strong knowledge of communication protocols such as OPC UA or MQTT, as well as cybersecurity practices to safeguard operational data, are further critical competencies for successful implementation.

4.5 Stage 5: IT Integration

A fundamental challenge in the adoption of AAS is the transition from a system-centric to an asset-centric information architecture. In a system-centric paradigm, asset-related data is stored and managed within the boundaries of individual enterprise systems, such as ERP, MES, or PLM, resulting in fragmented and siloed data landscapes. In contrast, an asset-centric architecture centralizes all asset-related data within a dedicated asset repository, where the asset itself becomes the authoritative source of its own information. Transitioning to this paradigm requires the systematic migration of asset-related data from each source system into the asset repository, while simultaneously establishing bidirectional interfaces that enable these systems to read and write asset data from the central repository rather than maintaining local copies. Since a single asset may hold relevant data across multiple enterprise systems, this integration must be performed for each affected system individually, rendering IT integration one of the most complex and resource-intensive steps in the implementation of an asset-centric data architecture.

The successful execution of this integration step demands a combination of architectural, technical, and data management competencies. An understanding of both system-centric and asset-centric information architectures is fundamental, complemented by expertise in enterprise system landscapes, including ERP, MES, and PLM environments. Technical proficiency in designing and implementing APIs, middleware, and bidirectional interfaces is essential to establish the required connectivity between source systems and the central asset repository. Furthermore, skills in data migration, including data extraction, transformation, and quality assurance, are critical to ensure a consistent and reliable transfer of asset-related data across multiple systems.

Stage	Description	Required Roles
S1	Infrastructure Groundwork	Solution Architect [26], DevOps Engineer [27], Cybersecurity Specialist [28]
S2	Asset Identification	Automation Engineer [29], Maintenance Technician [30]
S3	Basic Submodels	Data Engineer [31], Domain Expert
S4	Advanced Submodels	Data Engineer, Domain Expert, Network Engineer [32]
S5	IT Integration	Solution Architect, Software Engineer [33], Data Engineer

Table 1 STAMODIAAS stages and required roles

4.6 Use Case Mapping

This section presents six typical use cases relevant to production derived from [34]. All use cases require a digital representation of the production system as a fundamental prerequisite. To utilize the AAS for this purpose, the required infrastructure must first be established as a foundation. Depending on the type of data required and functionality, higher levels of integration become necessary.

Use Case 1 (UC1) addresses asset identification, such as listing all machines within a shop floor or locating a specific asset, for which S2 is sufficient. In contrast, Use Case 2 (UC2) involves retrieving more detailed machine information, such as a digital nameplate or maintenance report, to maintain the machine, which requires S3. Furthermore, Use Case 3 (UC3) focuses on technical monitoring. If live data is to be retrieved and displayed (e.g., PLC data) within a technical monitoring system, advanced submodels must be implemented, making S4 sufficient. Integrating value-

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based services, such as remaining useful life calculations or predictive maintenance into the production plant, corresponds to Use Case 4 (UC4). In this case, real-time data must be modeled and made accessible so that interfaces to external services can be created or used, which requires S4. A product-oriented use case, Use Case 5 (UC5), involves the automated collection of relevant information from the production process within a digital nameplate of the produced asset. Since this requires the availability of operational data within the plant's AAS as well as integration into a smart production system, including interfaces and data storage, we consider this use case to require S5. Comparable use cases, such as the automated generation of a carbon footprint, have similar requirements. Finally, Use Case 6 (UC6) builds on the other use cases and includes automated production workflows with a fully integrated MES for production orchestration or agent-based autonomous production. This scenario also requires S5, although in practice it demands additional development efforts beyond that level. **Table 2** provides an overview of the use cases and their mapping to the corresponding STAMODI-AAS stages.

Use Case	Description	Required Stage
UC1	Asset Inventory	S2
UC2	Maintenance Work	S3
UC3	Technical Monitoring	S4
UC4	Value-based Services	S4
UC5	Automated Digital Nameplate	S5
UC6	Automated Production System	S5

Table 2: Mapping of production use cases to STAMODI-AAS stages

5 Implementation

To validate STAMODIAAS, we implemented AAS-based production networks in two academic demonstrators. Building on these results, the proposed model was subsequently evaluated using real-world processes in collaboration with an industrial partner, enabling the derivation of practical results and recommendations.

5.1 Testbed for Software-Defined Value Network

At the Institute for Control Engineering of Machine Tools and Manufacturing Units (ISW) at the University of Stuttgart, a demonstrator for a Software-defined Value Network called *InnoWindow* has been developed (**Figure 2**). The demonstrator enables the evaluation of flexible cross-company production in an industry-scale environment [35]. Thereby, the implementation of the AAS was carried out by four PhD students with mechatronics and



computer science backgrounds, who regularly conduct Industry 4.0 projects and research on AAS. Also, they acted as domain experts with knowledge about the machines and to enterprise data engineers.

Figure 2: Software-defined Value Network of the ISW

In order to set up the AAS network, STAMODIAAS was applied as a structured rollout methodology. The objective was to incrementally transform the modular production environment into an asset-centric production network while systematically evaluating effort, competencies, and technical implications.

Following S1, Infrastructure, a distributed AAS backend based on Eclipse BaSyx[2] was deployed. Each production segment was equipped with an industrial edge PC hosting a containerized BaSyx setup, including AAS server, registry, and supporting middleware services. Additionally, a central AAS repository and registry were established for shell discovery and identifier resolution. The installation and configuration of the BaSyx environment for all five edge devices required software engineering expertise, particularly for container orchestration, service configuration, API exposure, and secure network integration, and was completed in one day, making this stage low effort. Distributed edge deployment increased initial setup complexity but improved modularity and scalability.

The second stage focused on the systematic cataloging of all relevant shop floor assets. For each module, a globally unique Asset-ID was defined and an empty AAS instance was instantiated. The technical instantiation of a single AAS required low effort, while clarifying asset boundaries within modular production cells required medium effort, as coordination between automation engineers and domain experts was necessary.

In S3, descriptive submodels such as Digital Nameplate, Technical Data, and Documentation were implemented. When structured source data was available, creating a single basic submodel required medium effort. Domain knowledge and AAS data modeling skills were the main competencies required.

Following S4, advanced submodels represented the most demanding modeling stage. Operational submodels, including condition monitoring and OPC UA integration, required the development of connector and semantic transformation logic due to missing direct mappings. This stage involved close collaboration between domain experts and

[2] Eclipse BaSyx: https://basyx.org

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software engineers and required high effort. The rollout confirmed that infrastructure setup required relatively low effort, while identification and implementation of advanced submodels demanded high effort.

5.2 Modular Production System

At the Institute for Production and Informatics (IPI) of Hochschule Kempten at the Technology Transfer Center Sonthofen, a transfer system demonstrator, as shown in **Figure 3**, is operated as a realistic shop floor environment for research and teaching. The demonstrator represents a modular production system with conveyor-based material transport, industrial control and handling components, and is used for investigations on real and virtual production. This makes it a suitable academic environment for evaluating the practical implementation of AAS-based concepts. The implementation was carried out in this context by contributors with practical experience in AAS modeling and implementation, complemented by practical expertise in operating and interacting with the shop floor system.

STAMODIAAS was applied to the transfer system as a structured rollout model for the AAS. Stages 1 to 4 were implemented step by step in the demonstrator in order to observe the associated technical effort and required competencies in a realistic shop floor setting.

Following Stage 1, Infrastructure, the AAS environment for the transfer system was set up using the Eclipse BaSyx. In addition, AASX Package Explorer[3] was used to model the AAS and its submodels. This stage established the technical basis for the subsequent implementation steps and enabled the demonstrator to host and manage the digital representation of the transfer system. The implementation can be regarded as low to medium effort, as the setup was limited to a single demonstrator environment.

In Stage 2, Identification, an AAS was created for the transfer system and assigned a unique identifier. To establish the physical-to-digital link, a QR code referring to the AAS identifier was attached to the transfer system. This stage enabled the demonstrator to represent the physical asset through a uniquely addressable digital counterpart. The technical implementation required low effort, while the correct definition of the asset identity and the physical linking of the QR code to the demonstrator required coordination between the digital model and the real shop floor asset.

In Stage 3, Basic Submodels, descriptive submodels such as Digital Nameplate, Technical Data, and Handover Documentation were implemented for the transfer system. These submodels were used to provide static asset-related information and formed the basis for a more structured digital representation of the demonstrator. Compared to the previous stages, the effort can be regarded as medium, since the required data first had to be identified and organized before being modeled in the AAS.

In Stage 4, Advanced Submodels, the digital representation of the transfer system was extended with live operational data. For this purpose, a Time Series Data (TSD) submodel, an Asset Interface Description (AID) submodel, and an Asset Interface Mapping Configuration (AIMC) submodel were implemented. The TSD submodel was deployed using Eclipse BaSyx with InfluxDB as the backend for storing and managing runtime data. The AID submodel



Figure 3: Demonstrator at IPI

was used to describe the OPC UA interface of the transfer system, including endpoint metadata and available datapoints. Based on this information, a client was developed to establish the connection to the OPC UA server and retrieve the relevant signal values. In accordance with the AIMC specification, the AIMC submodel defined the mapping of these datapoints to specific elements within the target AAS submodels. The client parsed both the AID and AIMC submodels to determine how retrieved data should be processed and where it should be written within the AAS. In contrast to the previous stages, this stage required stronger competencies in connectivity and IT/OT integration for configuring the interfaces, implementing the data mapping logic, and ensuring consistent data flow between the physical asset and its digital representation. Accordingly, the effort for this stage can be regarded as high compared to the implementation of the descriptive submodels.

Overall, the implementation at HS Kempten showed that the early STAMODIAAS stages could be realized with manageable effort in an academic demonstrator, while the implementation of advanced submodels and live data integration required significantly higher effort. In particular, the connection to the OPC UA server and the handling of time-series data demanded stronger competencies in connectivity and IT/OT integration. The rollout therefore confirmed that infrastructure setup and identification required comparatively low effort, whereas the transition to operational AAS use was the most demanding stage.

[3] https://github.com/eclipse-aaspe/package-explorer

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5.3 Validation in Automotive

As part of a validation initiative in the automotive industry, the AAS concept was implemented and tested in a demo setup in a paint shop environment. The implementation used Eclipse BaSyx to create and manage digital representations of physical assets along the paint shop line. Data from existing IT systems was transformed into submodels and added to the corresponding asset repository, making existing information accessible in a standardized way. The assets were additionally linked to a knowledge graph, enabling a richer and more connected representation of asset information and relationships. During the project, several challenges came up. One of the biggest challenges was identifying the right use cases and the appropriate submodels, since the large variety of assets and processes made it difficult to define a clear and meaningful scope. Another challenge was finding the correct roles and responsibilities. A third challenge was integrating an asset-centric architecture into an already existing system-centric landscape, where legacy systems and established data flows had to be connected and aligned with the AAS approach. These observations further highlight the importance of a structured, stage-based approach such as STAMODIAAS.

5.4 Results & Recommendations

The implementations in the academic demonstrators and the validation in the industrial context indicate that STAMODIAAS provides a useful structure for introducing AAS stage by stage in production environments. A central result is that the effort is not distributed evenly across the stages. Infrastructure setup and asset identification can usually be realized with comparatively low effort once a suitable AAS platform and deployment concept have been selected. In contrast, the transition from descriptive to operational submodels featuring dynamic data represents a clear increase in technical complexity, since it requires live data connectivity, interface configuration, semantic mapping, and the handling of changing information.

A second result concerns the role of available source data. Where structured and well-documented asset information already exists, the implementation of basic submodels such as Digital Nameplate, Technical Data, or Handover Documentation can be carried out with manageable effort. However, if information is incomplete, inconsistent, or distributed across multiple sources, the required effort rises significantly because data must first be identified, consolidated, and prepared for AAS modeling. This effect becomes even more pronounced in advanced submodels, where the quality and accessibility of operational data directly determine implementation complexity.

Based on these findings, a first recommendation is to start with clearly delimited use cases and a small set of relevant assets rather than aiming at broad AAS coverage from the outset. Early stages should focus on establishing a stable technical foundation, assigning unambiguous asset identities and implementing a limited number of descriptive submodels with immediate practical value. The use of standard submodels should be prioritized wherever possible, since they reduce modeling effort and improve interoperability.

A second recommendation is to treat advanced submodels and higher integration stages as dedicated engineering tasks rather than as simple extensions of descriptive AAS content. As soon as live data, interfaces, or event-driven behaviour are required, competencies in IT/OT integration, communication protocols, and operational data handling become critical. For this reason, implementations should plan sufficient effort for connector development, semantic mapping, and validation of the end-to-end data flow.

Finally, the results suggest that full IT integration should be considered a long-term transformation step rather than an initial rollout target. Integrating AAS into existing enterprise landscapes requires not only technical interfaces, but also architectural decisions regarding data ownership, governance, and the transition from system-centric to asset-centric information management. STAMODIAAS can therefore be applied as an incremental roadmap for practical AAS adoption: early stages enable quick and manageable entry points, while later stages provide orientation for scaling toward operational and enterprise-wide use.

6 Conclusion and Future Work

As many Industry 4.0 use cases require a shift from plant-centric to asset-centric automation, AASs can serve as a foundation for enhancing production systems. However, practical constraints as well as uncertainty and limited knowledge often hinder the implementation of operational AAS applications within companies. While current state-of-the-art research mainly focuses on innovative AAS use cases, an approach that considers the practical aspects of implementation is still missing.

In this paper, we propose STAMODIAAS, a five-stage model specifying the stages, roles and required effort for AAS integration in industrial production. The approach was applied in two academic demonstrators and validated in the production environment of the industrial partner AUDI, confirming the relevance of the proposed model.

Future work could investigate STAMODIAAS in a broader range of production environments across different industry sectors. Furthermore, a more detailed quantification of the required workload through additional practical experiments would help to further assess the resources needed to complete the integration stages and support decision makers in evaluating the implementation of AAS in their production systems.v Another aspect that will become increasingly relevant in the future is the relation between the AAS and dataspace technologies, as well as reference implementations such as Catena-X[4] or concepts like the M-X Port[5].

[4] https://catena-x.net/

[5] https://factory-x.org/wp-content/uploads/MX-Port-Concept-V1.10.pdf

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